

Calculus 2 - Homework 5 (Worked Solutions)

1.

Decompose $1/((x-1)(x+1)) = A/(x-1) + B/(x+1)$.

Clear denominators: $1 = A(x+1) + B(x-1)$.

$x = 1$: $1 = 2A \rightarrow A = 1/2$. $x = -1$: $1 = -2B \rightarrow B = -1/2$.

$\int \left[\frac{1/2}{(x-1)} - \frac{1/2}{(x+1)} \right] dx = \frac{1}{2} \ln|x-1| - \frac{1}{2} \ln|x+1| + C$
 $= \frac{1}{2} \ln \left| \frac{(x-1)}{(x+1)} \right| + C$.

ANSWER: $\frac{1}{2} \ln \left| \frac{(x-1)}{(x+1)} \right| + C$

2.

The denominator factors as $x(x+1)$, so decompose rather than substitute.

$1/(x(x+1)) = A/x + B/(x+1)$. $1 = A(x+1) + Bx$.

$x = 0$: $A = 1$. $x = -1$: $1 = -B \rightarrow B = -1$.

$\int \left[\frac{1}{x} - \frac{1}{(x+1)} \right] dx = \ln|x| - \ln|x+1| + C = \ln \left| \frac{x}{(x+1)} \right| + C$.

ANSWER: $\ln \left| \frac{x}{(x+1)} \right| + C$

3.

A comparison instinct is natural, but this integral evaluates exactly via substitution, which also settles convergence.

Let $u = \ln x$, $du = (1/x) dx$. When $x = 2$, $u = \ln 2$; as $x \rightarrow \infty$, $u \rightarrow \infty$.

The integral becomes $\int_{\ln 2}^{\infty} u^{-2} du$
 $= \lim_{s \rightarrow \infty} \left[-1/u \right]_{\ln 2}^s = \lim_{s \rightarrow \infty} (-1/s + 1/\ln 2) = 1/\ln 2$.

Converges to $1/\ln 2$.

ANSWER: converges to $1/\ln(2)$

4.

Odd power of cosine: peel off one cosine and convert the rest to sine.

$\cos^3 x = \cos^2 x \cdot \cos x = (1 - \sin^2 x) \cos x$.

Let $u = \sin x$, $du = \cos x dx$.

$\int (1 - u^2) du = u - u^3/3 + C$.

Back-substitute: $\sin x - \sin^3 x / 3 + C$.

ANSWER: $\sin(x) - \sin^3(x)/3 + C$

5.

Use integration by parts: $\int u dv = uv - \int v du$.

By LIATE choose $u = x$ (algebraic) and $dv = e^x dx$.

Then $du = dx$ and $v = e^x$.

So $\int x e^x dx = x e^x - \int e^x dx = x e^x - e^x + C$.

Factor: $(x - 1) e^x + C$.

Check: $d/dx[(x-1)e^x] = e^x + (x-1)e^x = x e^x$.

ANSWER: $(x - 1) e^x + C$

6.

The oscillating numerator can look like an alternating series, but $\sin^2(n)$ is always nonnegative, so this is a positive-term series suited to direct comparison.

For all n , $0 \leq \sin^2(n) \leq 1$, hence $0 \leq \sin^2(n)/n^2 \leq 1/n^2$.

$\sum 1/n^2$ converges ($p = 2$), so by Direct Comparison

$\sum \sin^2(n)/n^2$ converges (in fact absolutely).

ANSWER: converges (direct comparison with $\sum 1/n^2$)

7.

Irreducible quadratic factor $x^2 + 1$ needs a linear numerator:

$$(3x+1)/((x+1)(x^2+1)) = A/(x+1) + (Bx + C)/(x^2 + 1).$$

$$3x + 1 = A(x^2 + 1) + (Bx + C)(x + 1).$$

$$x = -1: -2 = A(2) \rightarrow A = -1.$$

$$\text{Expand and match: } (A + B)x^2 + (B + C)x + (A + C).$$

$$x^2: A + B = 0 \rightarrow B = 1. \quad x^1: B + C = 3 \rightarrow C = 2. \quad \text{const: } A + C = -1 + 2 = 1 \text{ (checks).}$$

$$\text{integral } [-1/(x+1) + (x + 2)/(x^2 + 1)] dx.$$

$$\text{integral } (x)/(x^2+1) dx = (1/2) \ln(x^2+1); \text{ integral } 2/(x^2+1) dx = 2 \arctan x.$$

$$\text{Result} = -\ln|x+1| + (1/2) \ln(x^2 + 1) + 2 \arctan x + C.$$

$$\text{ANSWER: } -\ln|x + 1| + (1/2) \ln(x^2 + 1) + 2 \arctan(x) + C$$

8.

The n^2 in the denominator might suggest a p-series comparison, but the exponential 2^n dominates, and the Ratio Test is decisive.

$$a_n = 2^n/n^2. \text{ Ratio:}$$

$$a_{n+1}/a_n = [2^{n+1}/(n+1)^2] * [n^2/2^n] = 2 * n^2/(n+1)^2.$$

$$L = \lim 2 * (n/(n+1))^2 = 2 * 1 = 2.$$

Since $L = 2 > 1$, the series diverges by the Ratio Test.

ANSWER: diverges (ratio test, $L = 2 > 1$)

9.

$$\Delta x = (2 - 0)/4 = 0.5. \text{ Nodes: } x_0=0, x_1=0.5, x_2=1.0, x_3=1.5, x_4=2.0.$$

$$f(x)=x^2 \text{ values: } f_0=0, f_1=0.25, f_2=1, f_3=2.25, f_4=4.$$

$$\text{Trapezoidal: } T = (\Delta x/2)[f_0 + 2f_1 + 2f_2 + 2f_3 + f_4].$$

$$T = (0.5/2)[0 + 2(0.25) + 2(1) + 2(2.25) + 4]$$

$$= 0.25[0 + 0.5 + 2 + 4.5 + 4] = 0.25(11) = 2.75.$$

(For reference the exact value is $8/3 = 2.6667$, so T_4 overestimates.)

$$\text{ANSWER: } T_4 = 2.75$$

10.

The denominator $x^2 + 4$ does not factor over the reals, so partial fractions is not the tool here. Instead notice the numerator is (up to a constant) the derivative of the denominator.

$$\text{Let } u = x^2 + 4, \text{ so } du = 2x dx, \text{ giving } x dx = (1/2) du.$$

$$\text{The integral becomes } \text{integral } (1/2)(1/u) du = (1/2) \ln|u| + C.$$

$$\text{Since } x^2 + 4 > 0 \text{ always, back-substitute: } (1/2) \ln(x^2 + 4) + C.$$

$$\text{ANSWER: } (1/2) \ln(x^2 + 4) + C$$